Conjecture 00000001002: a secant-count obstruction

lizaixi01, with OpenAI Codex assistance

23 October 2026

1 Statement and counterargument

Let K be any finite field, $q = |K|$, and let $t_2(2, q)$ be the least size of a complete arc in $\mathrm{PG}(2, q)$. An arc meets each projective line in at most two points. Completeness means that inserting any point outside it destroys this property. The conjecture asserts, in particular,

$$\liminf_{q \to \infty} \frac{t_2(2, q)}{\sqrt{q}} = 1,$$

where q ranges over all prime powers. We prove uniformly over every finite field that a complete arc of size k satisfies

$$k^2 > 2q, \qquad \frac{t_2(2, q)}{\sqrt{q}} \geq \sqrt{2} > 1.$$

Consequently the displayed liminf cannot equal one. This also rules out the stronger $\sqrt{q}+o(\sqrt{q})$ assertion.

2 The counting proof

Fix a complete arc S of size k. If $x \notin S$, then $S \cup \{x\}$ is not an arc. Some projective line therefore contains x and two distinct points of S: it cannot contain three points of S. Thus all projective points are covered by S and the secants indexed by its two-element subsets. The plane has $q^2 + q + 1$ points; each such line has $q + 1$ points. Overlaps only reduce a union's size, so

$$q^2 + q + 1 \leq k + (q + 1)\binom{k}{2}.$$

Multiplying by two and using $2\binom{k}{2} = k^2 - k$ gives

$$2(q^2 + q + 1) \leq (q + 1)k^2 - (q - 1)k.$$

Since $q \geq 2$ and $k \geq 0$, an assumption $k^2 \leq 2q$ makes the right side at most $2q(q + 1)$, a contradiction.

The definitions of arc and completeness agree with Bartoli et al., *Upper bounds on the smallest size of a complete arc in the plane PG(2,q)*, introduction. That paper records a stronger classical lower bound. The accompanying Lean proof independently proves the counting inequality and $k^2 > 2q$ above; it does not claim to formalize the stronger bound with an additional constant.

3 Actual finite geometry and the minimum in Lean

The point space is Mathlib's `Projectivization K (Fin 3 → K)`. Lines are the projective point sets of two-dimensional linear subspaces. The predicate `IsArc` bounds each such intersection by two, and `IsCompleteArc` forbids every outside insertion.

A maximum-cardinality arc exists in the finite collection of arc subsets. It is complete. A minimum-cardinality member of the resulting nonempty finite collection of complete arcs defines `minArc` and `minSize`. Their completeness and minimum specifications are proved. In particular the purported minimum is obtained from actual arcs, rather than assigned a numerical formula.

Mathlib's projective cardinality theorem gives the plane's exact size. To bound a line, `restrictPoint` sends an ambient projective point on a subspace to a projective point of that subspace. Mapping back recovers the original point and proves injectivity; the dimension-two cardinality theorem bounds its point count by $q+1$. For an outside point, failed insertion gives a line containing two old points. Uniqueness of the line through two distinct projective points identifies it with their span. The finite union over `powersetCard 2` now proves `cover_count`, and algebra proves `complete_arc_square_bound`.

4 The full asymptotic scope

The theorem `minRatio_lower` holds for every finite field, with no restriction to prime fields or to a selected subsequence. For any sequence of finite fields, the extended nonnegative real liminf of these ratios is at least $\sqrt{2}$. In particular it applies when the field sizes enumerate all prime powers in increasing order. The final real-valued theorem explicitly permits any field-size sequence tending to infinity and proves that its liminf is not one. Hence it covers the conjecture's full domain.

For precision about library conventions, real and nonnegative-real liminf are conditionally complete operations. Lean also handles their unbounded fallback branch, where the library value is zero and therefore not one. The geometric obstruction is the uniform lower bound and the separate extended-real liminf theorem; no fallback value is presented as a geometric example. Whenever the usual mathematical liminf is finite, the uniform lower bound forces it to be at least $\sqrt{2}$; an infinite liminf is likewise incompatible with the asserted value one.

5 Verification

The project pins Lean 4.33.0 and Mathlib commit

`db584cd6d46c92f209a44c0f1c829460d327499d`.

Source-bound receipts record an independent new Main build, the transitive axiom audit and official fresh kernel replay. No `sorry`, `native_decide` or additional axiom is used. OpenAI Codex assisted; a separate supervising Codex agent reviewed the complete formal chain. This is AI review, not human certification. Organizer acceptance is pending.